

Ashwin Gopalsamy

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SUMMARY

Staff Software Engineer at Visa building **payment infrastructure, distributed systems** and high-scale backend platforms. Own and build critical **Go** services processing **nearly a billion real-time payment interactions daily** across global payment networks, including multi-region infrastructure operating under PCI-DSS requirements. Published **machine-learning researcher** (Springer, 2022), MS in Artificial Intelligence student at **UT Austin** and member of Visa's **GenAI Ambassadors** driving AI adoption across engineering teams.

SKILLS

- **Languages:** Go (Golang) as primary language; working knowledge of Python, JavaScript and Bash.
- **Architecture:** CQRS, Event Sourcing, Event-Driven Architecture, Hexagonal and Multi-Region Distributed Systems.
- **Payments & Banking:** ISO 8583, card authorization, payment rails, core banking, interest engine and card-network protocols across Visa, Mastercard, RuPay, etc.
- **Cloud & Platform:** AWS, Kubernetes, Terraform, Docker, Argo CD Codefresh and Concourse CI/CD.
- **Data & Storage:** PostgreSQL, MariaDB, DynamoDB, Cassandra, Redis, Snowflake and Athena.
- **Reliability, Observability & Security:** Grafana, Prometheus, Sentry, structured logging, SonarQube, Checkmarx SAST, PCI-DSS and PSD2.

WORK EXPERIENCE

Staff Software Engineer, Pismo, a Visa company.

June 2025 - Present - Remote.

- Own Pismo's ISO 8583 authorization domain, a Go, AWS and Kubernetes stack processing **nearly a billion payment interactions daily** across Visa, Mastercard, RuPay, Tecban, Alto and Elo networks.
- Scaling our cloud-native authorization platform powering banks and fintechs globally, delivering real-time authorization decisioning at greater than **99.995% availability**.
- Building PCI-DSS 4.0 compliant, high-availability systems across a **multi-region zero-downtime** setups, aligned with Visa's global-grade engineering practices.
- Engineering cross-region authorisation infrastructure that interacts with cryptographic hardware modules in real-time ensuring seamless failover capabilities, platform integrity, modularity and long-term resilience.
- Led evaluation and adoption of AI-assisted development tooling as part of Visa's **Generative AI Platform**, establishing AI-native coding standards, cross-tooling integrations and workflows, across **engineering teams** and authoring AI development guidelines adopted by **org-wide** at Pismo.

Professional Engineer, Solaris SE.

June 2022 - June 2025 - Remote.

- Engineered the complete lifecycle of 'Credit Card Interest' product, from design to post-release, collaborating with internal teams and product owners, generating revenue over **EUR 30M annually**.
- Owned 'Accounts Domain', with AWS, Kubernetes, Terraform stack supporting over **20M transactions per day**.
- Engineered 100% cloud-native, CQRS event-sourced Core-banking systems, achieving **99.999% uptime**.
- Led end-to-end development of 'Savings Account' product, driving **significant revenue growth**.
- Maintained on-call rotations, observability dashboards and runbooks that improved **incident response times**.

Associate Software Engineer, Hexaware Technologies.

June 2021 - June 2022 - Remote.

- Contributed to enterprise-level distributed applications, implementing solutions that improved system performance with project timelines met ahead of schedule.
- Deployed a Cloud HRMS product for a major automotive client, ensuring operation and user experience for over **75K employees** within a system valued at **USD 1.5B**.

Intern, Microsoft.

4 months - Remote.

- Developed features for an open-source project using Microsoft Azure Cloud and GitHub as part of the 'Future Ready Talent' program.

EDUCATION

The University of Texas at Austin

Fall 2026

Master of Science in Artificial Intelligence

Incoming

Coursework: Ethics in AI, Case Studies in ML, AI in Healthcare (*ongoing*); Machine Learning, Deep Learning, NLP, Reinforcement Learning, Advances in Deep Learning, Deep Generative Models, Planning/Search/Reasoning Under Uncertainty.

Hindusthan College of Engineering and Technology (Anna University)

August 2017 – May 2021

Bachelor of Technology in Information Technology

CGPA: 8.32/10.0 (Top 5%)

Coursework: Data Structures & Algorithms, Algorithm Design & Analysis, Artificial Intelligence, Discrete Mathematics, Probability & Statistics, Linear Algebra, DBMS, Operating Systems, Computer Networks, and Software Engineering.

Highlights: Graduated First Class; ranked #1 coder on Skillrack competitions for 10 consecutive weeks; active in student clubs and technical workshops.

RESEARCH PUBLICATION

Intelligent Traffic Management System Using Machine Learning | *Springer LNEE*, Vol. 905, 2022

- Designed and implemented a real-time vehicle detection system with dynamic signal adjustment based on live traffic density. Calibrated confidence thresholds for object detection under variable lighting conditions, addressing precision-recall tradeoffs in latency-sensitive environments.

CERTIFICATIONS

AWS Certified: Solutions Architect Associate | *Amazon Web Services*

AWS Certified: Cloud Practitioner | *Amazon Web Services*

Microsoft Azure Cloud: 3x Certified | *Microsoft Azure*